## SECTION-D

**Note:** Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

Q.23 Explain the construction, working, advantages and disadvantages of CSTR along with diagram. (CO4)

Q.24 Explain the construction, working, advantages and disadvantages of vapour absorption refrigeration cycle with the help of diagram. (CO3)

Q.25 Write short notes on the following (CO4)

a) Order of reaction and molecularity.

b) Homogeneous and Heterogeneous reactions.



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### 3rd Sem / Chemical

### Subject : Chemical Thermodynamics and Reaction Engineering

Time : 3 Hrs. M.M. : 60

## SECTION-A

**Note:** Multiple choice questions. All questions are compulsory (6x1=6)

Q.1 Pressure is a (CO1)

a) Extensive Property b) Intensive Property

c) Both a & b d) None of these

Q.2 Work done in a free Expansion Process is (Co2)

a) Positive b) Negative

c) Zero d) Maximum

Q.3 Which of the following laws deals with the concept of thermal equilibrium? (CO1)

a) Zeroth Law b) First Law

c) Second Law d) Third Law

Q.4 Gibb's free energy is defined as the maximum work obtained from a thermodynamic system at constant (CO1)

a) Temperature b) Pressure

c) Both A & B d) None of these

Q.5 Which of the following is not a flow reactor? (CO4)

a) PFR b) Batch Reactor

c) CSTR d) None of these

Q.6 Molecularity of a reaction can never be (CO4)

a) One b) Two

c) Three d) Zero

## SECTION-B

**Note:** Objective/ Completion type questions. All questions are compulsory. (6x1=6)

Q.7 Define State Function with Example. (CO1)

Q.8 Name any two commonly used refrigerants. (CO3)

Q.9 Define Enthalpy. (CO1)

Q.10 Define adiabatic process. (Co2)

Q.11 Define rate of reaction. (CO4)

Q.12 Name any two factors which affect the rate of reaction. (CO4)



## SECTION-C

**Note:** Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

Q.13 Explain the limitations of first law of thermodynamics. (CO2)

Q.14 Explain system, surrounding and boundry w.r.t chemical thermodynamics. (CO1)

Q.15 Explain Heat pump and its coefficient of performance. (CO3)

Q.16 Explain intensive properties and extensive properties with example. (CO3)

Q.17 Derive the relation between Cp and Cv for ideal gas. (CO2)

Q.18 Differentiate between endothermic and exothermic reactions. (CO4)

Q.19 Explain the affect of temperature and concentration on chemical equilibrium of a chemical reaction. (CO4)

Q.20 Explain the initial rate method for determining the order of reaction. (CO4)

Q.21 Explain the concept of activation energy in brief. (CO4)

Q.22 Explain the second law of thermodynamics in brief. (CO3)